More About Mutations: changing DNA

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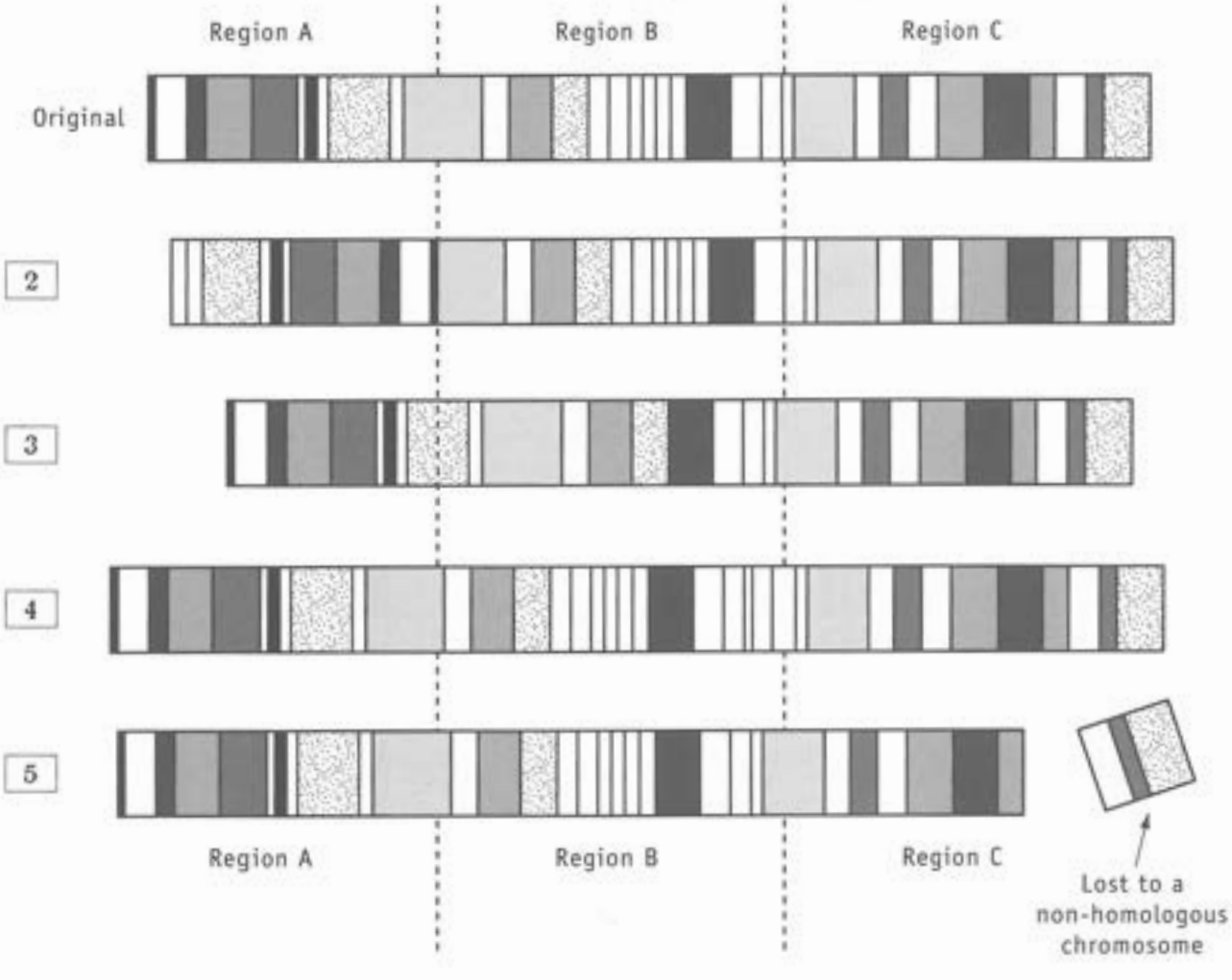
Throughout evolution, random changes have occurred to DNA, These changes can be harmful or beneficial, depending on how, and in what sort of environment, they are expressed. For example consider the following scenario. A species of bird has toes. Among this population, through mutation, some birds carry a gene producing webbed feet. Possession of webbed feet would improve swimming ability and ability to walk through marshlands. This would enable these birds to be more successful feeders in wet areas. The webbed-feet birds would breed successfully in wetlands and the trait would increase through time.

1 In what ways might the possession of webbed feet be a disadvantage for these birds?

Alternatively, a mutation might debilitate (weaken) an organism. For example, the ability to produce a vital digestive organism may be reduced or prevented. This would be an example of a harmful mutation. There are four main mechanisms involved in mutation:

- *deletion*, the removal and loss of genetic material
- *addition/duplication*, adding extra genetic material onto a chromosome. When this occurs between homologous pairs (the most common occurrence) it causes the duplication of genes already present on the chromosome.
- *translocation*, the movement of genetic material from one chromosome onto another non-homologous chromosome
- *inversion*, inverting information on a gene. This occurs when a chromosome breaks and reforms, but with the gene sequence inverted.

Compare the strands of DNA in examples 2–5 below with the original provided. Name the type of mutation that has occurred each time. Indicate the region (A, B or C) in which the mutation occurred.



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